



Sensitivity to elevated salinity in coastal dune plants

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Abstract

Coastal dune plants provide critical ecosystem and cultural services, which are severely threatened by ongoing climate change and sea-level rise. Coastal habitats account for a disproportionate extent of land on islands, emphasizing the importance of these ecosystems for island biodiversity. We investigated salinity tolerance in a diverse pool of 19 native and invasive plant species from Hawai‘i’s coastal dunes for insights into their vulnerability to sea-level rise. Salinity tolerance was investigated experimentally in seedling and juveniles by treating plants with artificial seawater for 3 weeks, followed by a 2-week freshwater recovery period. For mechanistic insights, stomatal conductance and leaf chlorophyll content were measured before, during, and after seawater treatment. Salinity tolerance was highly variable among species. The least tolerant species experienced 100% mortality and up to 95% reductions in growth in seawater compared to freshwater conditions. The most tolerant species grew more in seawater, consistent with a halophytic strategy. Most species had intermediate salinity tolerance, with generally high survival and variable reductions in growth under seawater treatment. Phenotypic plasticity in stomatal conductance and chlorophyll content was widespread, and species that maintained the highest levels of stomatal conductance and leaf chlorophyll content under seawater had the greatest biomass tolerance. Despite growing in close proximity to the ocean, Hawai‘i’s coastal dune plants are highly variable in salinity tolerance, and sea-level rise is likely to be a major threat, leading to reduced seedling establishment and population declines in some species, with cascading effects on associating biodiversity, emphasizing the need for urgent conservation actions.

Keywords Salt tolerance · Island ecology · Stomatal conductance · Chlorophyll · Sea-level rise

Introduction

Coastal ecosystems are globally important, providing critical ecosystem and cultural services (Mehvar et al. 2018). Driven by complex geomorphology and steep climate gradients associated with close proximity to the ocean, there is a diversity of habitat types and biological communities in coastal regions. Coastal dune, also known as sandy beach and coastal strand, ecosystems stand out as particularly

important by providing cultural resources, including recreation, ceremonies, and resource provisioning (Everard et al. 2010; UNEP 2006). Plants feature prominently within coastal dunes, stabilizing shore lines, preventing erosion, providing food, shelter, and other cultural resources, and fostering associated biodiversity via nesting sites for seabirds, turtles, and mammals. The persistence of coastal dune plants amidst rapid global change is thus an urgent research priority for conservation.

Coastal ecosystems worldwide are under threat from climate change as a result of sea-level rise, increasing aridity, and more frequent extreme events such as storms and flooding (Feagin et al. 2005; Masson-Delmotte et al. 2021). Coastal dunes typically lie above freshwater lenses fed by groundwater and precipitation (Stuyfzand 2017), and dune plants thus primarily rely on freshwater to function and grow (Greaver and Sternberg 2007). However, periodic seawater intrusion due to storms, salt spray, and flooding means that coastal dune plants may sometimes take up seawater

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(Greaver and Sternberg 2007). Increased storm frequencies and sea-level rise are projected to increase seawater exposure in coastal dunes, leading to elevated salinity stress for dune plants (Hanley et al. 2019). Exacerbating these increases in salt input to coastal dunes, reduced precipitation under climate change (Masson-Delmotte et al. 2021) contracts the freshwater lens, facilitating seawater intrusion and promoting the accumulation of salt ions in dune soils (Greaver and Sternberg 2010). Determining whether coastal dune plants can tolerate increasing salinity under climate change is critical for their conservation and is an urgent research priority.

Salinity causes direct and indirect stress to plants via effects on cellular function (membrane instability, interference with ion transport, and protein denaturation) and water balance (osmotic effects) (Munns 2002). Chronic exposure to high salinity is associated with the evolution of salinity tolerance, which generally involves mechanisms to avoid, sequester, or secrete salt ions (Flowers and Colmer 2015). Because of the increasing salinization of agricultural landscapes, salinity tolerance has been extensively investigated in crops and their relatives (Munns and Tester 2008). In contrast, salinity tolerance in native coastal dune plants has generally been examined as case studies, involving one or a few species for a given site (Ben Amor et al. 2006; Suarez 2011; Zunzunegui et al. 2021). Perhaps, this limited research effort reflects the general expectation that dune plants express at least partial salinity tolerance, if they are not fully halophytic, assumed to have evolved as a consequence of their close proximity to the ocean (Hesp 1991). Yet, without evidence for more species, it is impossible to predict how coastal dune community composition may shift under climate change. Species variability in salinity tolerance may lead to losses of intolerant species, while those with greater tolerance will likely increase in abundance. Because ecosystem and cultural services increase under high biodiversity (Everard et al. 2010; UNEP 2006), such community composition shifts will likely jeopardize coastal dune stability and service provisioning. A comparative approach that investigates salinity tolerance in co-occurring dune species is needed for a better understanding of community and ecosystem function as salinity increases due to climate change.

Oceanic islands are particularly threatened by sea-level rise, as coastal regions account for a disproportionate extent of land area. Moreover, island biodiversity is unique, with exceptionally high rates of species endemism (Fernández-Palacios et al. 2021; Tershy et al. 2015). Sea-level rise is an immediate threat to islands, with ongoing sea-level rise and seawater intrusion already compromising island-wide freshwater reservoirs in some low-lying, small islands (Doorga et al. 2024). While salinity tolerance has been examined in some islands near continents, particularly in Europe and North America (Belkheiri and Mulas 2013; van Puijenbroek

et al. 2017), relatively few studies have investigated salinity tolerance in Pacific Island species.

The Hawaiian Islands offer compelling opportunities to investigate climate change effects on the world's most isolated biota. Featuring exceptionally high endemism, > 90% for angiosperms (Wagner et al. 2023), considerable research has investigated Hawai'i's native plant vulnerability to global climate change (Alison and Donald 2024; Fortini et al. 2013; Westerband et al. 2020). However, much of this work has focused on forest species (Barton et al. 2021), which dominate the terrestrial landscape across the Hawaiian Islands. In contrast, less is known about Hawai'i's coastal plants, an important oversight considering that coastal dunes are among the most native-dominated biological communities in the heavily human-altered lowland habitats (Warshauer et al. 2009).

In Hawai'i, climate change has already been detected through declines in mean annual rainfall (Frazier and Giambelluca 2017), increases in the duration and magnitude of drought (Frazier et al. 2022), and more extreme precipitation events (Chu et al. 2010), over the past 100 years. These patterns are projected to continue or escalate in Hawai'i under future climate change (Timm et al. 2015; Xue et al. 2020), which, in combination with continued sea-level rise and more frequent coastal flooding (Thompson et al. 2019), will lead to increasingly high salinity in coastal dune habitats. As a result of these changing conditions, coastal plants face increasing salinity and drought stress, along with habitat loss due to beach erosion (Anderson et al. 2015; Kane et al. 2014; Romine et al. 2016). Salinity tolerance has been previously examined in three widespread, common Hawaiian dune species, all of which were found to be sensitive to elevated salinity, consistent with a glycophytic functional strategy (Goldstein et al. 1996; Lum and Barton 2020). Moreover, these species demonstrated particularly low salinity tolerance at the seed and seedling stages, as evidenced by reduced and delayed germination, low survival, and reduced photosynthetic activity and growth rates (Alpha et al. 1996; Lum and Barton 2020). A more comprehensive investigation into seed salinity tolerance revealed extensive variability among species, ranging from high tolerance consistent with a halophytic strategy, to intermediate, and even low salinity tolerance associated with high rates of seed mortality (Walsh et al. 2023). If germination bottlenecks influence community dynamics in Hawai'i's coastal dunes, as expected from other systems due to the vulnerability of early life stages to abiotic stress (Del Vecchio et al. 2020; Harper 1977), such variability in seed salinity tolerance is likely to drive shifts in community composition under future sea-level rise.

Following germination, seedlings must establish and grow, and so to expand our understanding of recruitment dynamics in Hawai'i's coastal dunes under sea-level rise, in this study, we tested salinity tolerance at the seedling and

juvenile stages across a phylogenetically diverse pool of 19 plant species from 15 plant families (Table 1). In addition to 14 native plant species (represented by 2 endemic and 12 indigenous species), we tested salinity tolerance in 5 non-native, invasive plant species (“invasive” hereafter) to assess the likelihood that sea-level rise may facilitate the spread of invasive species if they are more tolerant than native species (Morais et al. 2012). Because widespread species commonly harbor extensive genetic variation, the conservation of which is a priority under climate change (Sgrò et al. 2011), we tested salinity tolerance for multiple populations in two of the most common dune species, *Chenopodium oahuense* (Amaranthaceae) and *Gossypium tomentosum* (Malvaceae). We predicted that species would vary in salinity tolerance, with most aligning with the glycophytic strategy (i.e., sensitive to salinity stress), as has been previously observed for other coastal dune plants in Hawai‘i (Alpha et al. 1996; Goldstein et al. 1996; Lum and Barton 2020). Moreover, we predicted that seedlings would be more sensitive to elevated salinity than juveniles, consistent with general increases in stress tolerance across ontogeny (Barton 2024).

Using experimental approaches, we treated seedling and juvenile plants with artificial seawater, and quantified tolerance with respect to measures of fitness, namely survival and growth (Negrão et al. 2016). In addition, for mechanistic

insights into likely species variability in salinity tolerance, we quantified photosynthetic metrics throughout the experiment. Specifically, the regulation of stomatal conductance rates (and associated water use efficiency) and maintenance of high leaf chlorophyll content were previously identified to be significantly linked to salinity tolerance in Hawaiian dune plants (Lum and Barton 2020), and we predicted that species with greater salinity tolerance would similarly express these strategies in the species tested here. This study offers the first community-scale investigation of seedling salinity tolerance in an island flora threatened by sea-level rise, providing new insights into the evolutionary ecology of island plants while also contributing valuable information for their conservation.

Materials and methods

Species were selected on the basis of their abundance and widespread distribution, so that the sample pool includes the foundational and associating species of Hawai‘i’s coastal dune communities (Warshauer et al. 2009). Most tested species are woody, often prostrate and creeping shrubs, although a graminoid and a couple forbs were also included (Table 1). Experimental plants were grown from wild-collected seeds

Table 1 Characteristics of 19 coastal dune species included in salinity tolerance experiments. Two species were represented by multiple populations

Species	Family	Habit	Origin	Source population	Stage tested
<i>Bidens pilosa</i>	Asteraceae	Forb	Invasive	Maui	Both
<i>Casuarina equisetifolia</i>	Casuarinaceae	Tree	Invasive	Maui	Both
<i>Chenopodium oahuense</i>	Amaranthaceae	Tree	Endemic	3 populations: Kaua‘i, Maui, unknown	Seedling, seedling, both
<i>Cordia subcordata</i>	Boraginaceae	Tree	Indigenous	N/A	Both
<i>Fimbristylis cymosa</i>	Cyperaceae	Graminoid	Indigenous	Kaua‘i	Both
<i>Gossypium tomentosum</i>	Malvaceae	Shrub	Endemic	2 populations: Maui, Lāna‘i	Seedling, both
<i>Heteropogon contortus</i>	Poaceae	Graminoid	Indigenous	Maui	Both
<i>Ipomoea indica</i>	Convolvulaceae	Vine	Indigenous	Maui	Both
<i>Leucaena leucocephala</i> subsp. <i>leucocephala</i>	Fabaceae	Tree	Invasive	O‘ahu	Both
<i>Lycium sandwicense</i>	Solanaceae	Shrub	Indigenous	Kaua‘i	Seedling
<i>Pluchea indica</i>	Asteraceae	Shrub	Invasive	N/A	Seedling
<i>Plumbago zeylanica</i>	Plumbaginaceae	Shrub	Indigenous	Maui	Both
<i>Prosopis pallida</i>	Fabaceae	Tree	Non-native	Maui	Both
<i>Sesuvium portulacastrum</i>	Aizoaceae	Forb	Indigenous	Maui	Juvenile
<i>Thespesia populnea</i>	Malvaceae	Tree	Indigenous	Maui	Both
<i>Tribulus cistoides</i>	Zygophyllaceae	Vine	Indigenous	Maui	Both
<i>Vigna marina</i>	Fabaceae	Forb	Indigenous	Kaua‘i	Both
<i>Vitex rotundifolia</i>	Lamiaceae	Shrub	Indigenous	Maui	Both
<i>Waltheria indica</i>	Malvaceae	Shrub	Indigenous	Maui	Both

Habit = forb, graminoid, shrub, tree, and vine. Origin = endemic, indigenous, invasive. Seeds were collected from wild source populations across Maui, Kaua‘i, Lāna‘i, and O‘ahu. Most species were tested at both the seedling and juvenile stage, but in a few cases, low germination resulted in only a single stage being tested

from populations occurring across the Islands of Kauaʻi, Maui, Oʻahu, and Lānaʻi (Table 1).

Salinity tolerance was experimentally tested in the University of Hawaiʻi at Mānoa Pope Greenhouses in 2020–2022. Due to the large number of focal species and the labor-intensive collection of photosynthetic data, it was not logistically feasible to test all species simultaneously. Instead, species were tested sequentially across the 2-year experimental period. Because the UH-Mānoa Greenhouses are located in low-elevation Oʻahu, there is minimal seasonality, leading to relatively consistent temperatures and daylength throughout the year (Giambelluca et al. 2014). Nonetheless, in light of potential unaccounted for variability in conditions, we consider each species to be a separate trial, and we synthesize the treatment responses across species using meta-analysis. This is a conservative approach that enables us to assess general patterns across species, and is a recommended application of meta-analysis to analyze effect sizes calculated at the scale of species or sites within studies (Gurevitch 2013). Such use of meta-analysis to analyze patterns across species within studies is increasingly common in ecology (Barton and Shiels 2020; Fajardo and Siefert 2018; Nakagawa and Cuthill 2007).

To facilitate germination, seeds were scarified as needed to break physical dormancy (Lilleeng-Rosenberger 2005), and were sown in trays filled with Promix BX (65–75% Canadian sphagnum peat moss, perlite, dolomitic and calcitic limestone, macro- and micronutrients, *Glomus intraradices* mycorrhizae inoculum). Following germination, seedlings with 1 to 3 true leaves were transplanted into 1-gallon pots filled with mixed media of approximately 1:1:1 by volume Promix BX, black cinder, and coral sand, which resembles the coastal dune substrate in Hawaiʻi. Each plant received a single application of slow-release fertilizer (Osmocote) upon transplant, and seedlings were watered daily with fresh tap water to minimize transplant shock.

Plants were randomly assigned to experimental groups for 2 treatment factors, seawater and ontogeny. The seawater factor had 2 levels: the seawater treatment group, which received 3 weeks of artificial seawater, and the freshwater control group, which was watered throughout the experiment with fresh tap water. The ontogeny factor also had 2 levels: the seedling stage treated with seawater at 4–6 true leaves, and the juvenile stage treated at 10–12 true leaves. During the seawater treatment, plants were watered 3 times per week with 300 mL artificial seawater; control plants received the same volume of water during this period. Artificial seawater was prepared with Instant Ocean® marine salts to a concentration of 31.377 ppt, which approximates Hawaiʻi's natural seawater salinity (Walsh et al. 2023). To mimic reductions in salinity expected for coastal environments due to precipitation and fresh groundwater mixing, seawater treatments were then diluted to 20 ppt using a salinity probe (Pro 30

conductivity meter, YSI). Freshwater in the greenhouse was measured at 0.6 ppt using the salinity probe, confirming low salinity in the control freshwater group. The two experimental factors were crossed to produce a full factorial design, including seawater treatments applied at the seedling and juvenile stages, and control groups at the seedling and juvenile stages. Following the 3-week seawater treatment, all plants were watered with fresh tap water for 2 weeks to allow for a recovery, and then, control and seawater groups were harvested. Shoot and root tissues were extracted from the pots, oven-dried at 60°C for at least 7 days, and weighed on an analytical balance. The target experimental design included ten replicate plants per 2 × 2 experimental group per species, but due to mortality and variable germination rates, sample sizes varied across species, with a final total sample size of $N = 684$ (Supplementary Table S1). Unbalanced replication and missing ontogenetic stages in a couple species are easily accommodated using meta-analysis, providing additional justification for this statistical approach.

For insights into ecophysiological mechanisms associated with salinity tolerance, we collected photosynthetic data of seawater-treated and control plants. Leaf chlorophyll content (unitless measure of greenness) was measured with an SPAD meter (502Plus, Konica Minolta), and stomatal conductance ($\mu\text{mol m}^{-2} \text{s}^{-1}$) was measured with a porometer (Decagon Devices, Inc.). Trait data were collected before the onset of seawater treatment, at the end of the 3-week seawater treatment period, and again following the 2-week freshwater recovery period before plants were harvested.

Data were analyzed in R version 4.1.1 and RStudio version 2021.09.0. Species responses were calculated as effect sizes for seawater treatment effects on response variables (biomass, stomatal conductance, and chlorophyll) at each ontogenetic stage within species as log-response ratios: $\ln(M_S/M_C)$, where M_S is the mean response value of seawater treatment groups, and M_C is the mean response value of the freshwater control groups. Positive log-response ratios indicate higher growth or photosynthetic performance under seawater than control conditions, as would be predicted for halophytes, and negative log-response ratios indicate reduced growth or photosynthetic function under seawater compared to freshwater conditions, consistent with low salinity tolerance. For survival, which is a binary response variable, the odds ratio effect size was calculated as: $(A_S/D_S)/(A_C/D_C)$ where A represents the number of surviving plants while D represents those that died in the seawater-treated (S) and control (C) groups. Effect sizes calculated at the species scale were the unit of replication used in linear models (described below), allowing us to examine patterns across species.

Because our species pool was phylogenetically diverse, we investigated whether a phylogenetic signal was detectable for the 3 continuous response variables analyzed (LRR

for total biomass, stomatal conductance, and chlorophyll content). A tree of our focal species was pruned from a megacalibrated time-tree (Magallón et al. 2015), and used to calculate Blomberg's K and Pagel's λ with the *phylosig* function in the 'phytools' package (Revell 2012). Blomberg's K and Pagel's λ compare a given phylogeny to one expected under a Brownian model of evolution (Blomberg and Garland 2002; Münkemüller et al. 2012; Pagel 1999), and values close to zero indicate the lack of phylogenetic signal. Effect sizes were found to have a weak phylogenetic signal using both Pagel's λ and Blomberg's K ($P > 0.22$, Supplementary Table S2), and so, phylogeny was not incorporated into the final models.

Data were analyzed using the 'metafor' package, and standard deviations and sample sizes of the treatment and control groups were used for inverse-weighted estimation (Viechtbauer 2010). Mixed-effects models were conducted for each response variable to test whether responses differed between native and invasive species and between seedling and juvenile stages and their interaction using the *rma.mv* function. Because chlorophyll and stomatal conductance were measured repeatedly within species, these models included the additional factor of time, and to account for the repeated-measures design, species was included as a random factor. The significance of each factor was determined using likelihood ratio tests, which produces a test statistic with a Chi-square distribution. Model-generated group means and 95% confidence intervals are presented in the figures. When confidence intervals do not overlap with zero, the group can be concluded to have a significant effect of salinity. To facilitate interpretation, group mean log-response ratios are back-transformed to % difference between control and seawater treatment groups as: $100 - 100e^{(\text{response ratio})}$.

To investigate relationships between photosynthetic traits and seawater biomass tolerance (estimated as the log-response ratio of total biomass), linear regression analyses were conducted. Plasticity in chlorophyll and conductance after seawater treatment and following recovery are estimated at the species level with the log-response ratios, and for models in which the traits significant predict biomass responses indicate a mechanistic link between plasticity in photosynthesis and salinity tolerance. To investigate the potential for pre-seawater photosynthetic function in tolerance, the mean trait values before seawater treatment were also tested in linear regression models with biomass LRR.

Results

Survival was high in most species (Supplementary Table S3), with no detectable difference between seedling and juvenile stages ($\chi^2 = 0.19$, $P = 0.6654$) or between native and invasive species ($\chi^2 = 2.22$, $P = 0.1367$) in salinity

effects on survival. While most species had high survival under both freshwater and seawater conditions (Supplementary Table S3), four species suffered very high mortality ($> 75\%$) under seawater treatment at both the juvenile and seedling stages: the native species *Ipomoea indica*, *Plumbago zeylanica*, and *Waltheria indica*, and the invasive *Bidens pilosa*. In addition, the native *Vitex rotundifolia* and *Cordia subcordata* suffered nearly 100% mortality under seawater at the seedling stage, while demonstrating almost no mortality at the juvenile stage, indicating an increase in salinity tolerance across ontogeny.

Seawater negatively affected plant growth, measured as total biomass, significantly more at the seedling than juvenile stage ($\chi^2 = 3.90$, $P = 0.0484$, Fig. 1). We did not detect a significant difference in seawater effects on total biomass between native and invasive species ($\chi^2 = 2.21$, $P = 0.1373$), nor was there evidence for an interaction between species origin and ontogenetic stage ($\chi^2 = 1.23$, $P = 0.2670$). Nonetheless, it seems that native species in particular achieve an increase in salinity tolerance as they mature from the seedling to juvenile stage, while introduced species have persistently low tolerance throughout early ontogeny (Fig. 1).

Photosynthetic traits demonstrated significant phenotypic plasticity in response to seawater treatments, although the magnitude of effect depended on the sampling time (Table 2). As expected, no differences between control and treatment groups were detected before seawater was administered (Pre-seawater effects in Figs. 2 and 3). Immediately following the 3-week seawater treatment, leaf chlorophyll content was approximately 10% lower in treatment than control plants (Fig. 2), while stomatal conductance declined by nearly 70% in seawater plants (Fig. 3). The freshwater recovery period enabled plants to partially regain stomatal conductance and leaf chlorophyll content, although the extent of recovery was greater in juvenile than seedling stages (Figs. 2, 3), leading to a significant Stage $\times$ Time interaction (Table 2).

Salinity tolerance estimated as total biomass LRR varied considerably among species (Supplementary Fig. S2), and was significantly linked to plasticity in photosynthetic traits (Table 3). In both ontogenetic stages, plants achieved greater tolerance in total biomass in species that minimized reductions in leaf chlorophyll content and stomatal conductance (Fig. 4). In contrast, those species expressing the greatest declines in leaf chlorophyll content and stomatal conductance had the greatest reductions in growth under seawater treatment (Fig. 4). This trend continued through the freshwater recovery period, with higher tolerance linked to greater recovery of leaf chlorophyll content, particularly in seedlings (Table 3, Supplementary Materials Fig. S3).

Fig. 1 Mean effect sizes for seawater effects on growth for seedling and juvenile stages of native and invasive coastal dune species. Percent change in total biomass is calculated from log-response ratios, with bias-corrected 95% confidence intervals. Negative values indicate reduced growth in seawater-treated compared to control plants receiving freshwater throughout the experiment. Effect sizes with 95% confidence intervals that do not overlap with zero are significant

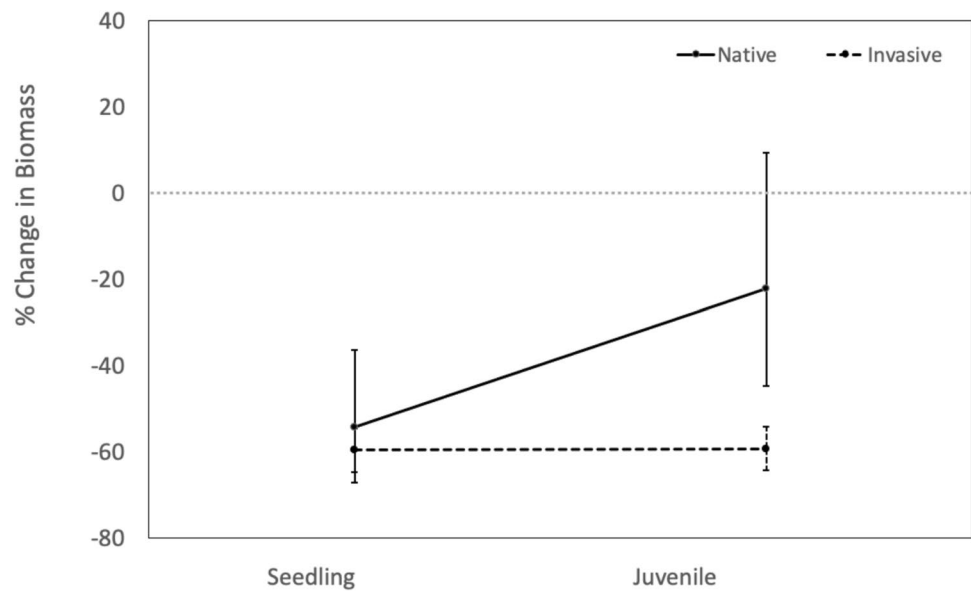


Table 2 Variability in seawater effects on plant functional traits for species origin (native vs. invasive), plant ontogenetic stage (seedling vs. juvenile), and sampling time (before seawater treatment, after seawater treatment, following freshwater recovery period)

Trait	Origin	Stage	Time	Stage x time
Chlorophyll	0.25	0.73	132.61 ***	13.55 **
Conductance	0.08	8.79 **	1086.10 ***	39.13 ***

Likelihood ratio tests are reported as Chi-square values, and significance is given as *** ($P < 0.0001$), ** ($P < 0.01$), * ($P < 0.05$), + ($P < 0.10$)

Discussion

Coastal dune plants in Hawai'i vary considerably in salinity tolerance, indicating a wide range of vulnerability to sea-level rise under climate change. Using experimental approaches, we determined that most Hawaiian dune plant species are glycophytes, plants that express sensitivity to seawater exposure associated with reduced photosynthetic activity and growth. One notable exception is *Sesuvium portulacastrum*, which grew more under seawater than

Fig. 2 Mean effect sizes for seawater effects on leaf chlorophyll content for seedling and juvenile stages assessed before seawater treatment, after 2 weeks of seawater treatment, and following 3 weeks of freshwater recovery. Percent change in chlorophyll is calculated from log-response ratios, with bias-corrected 95% confidence intervals. Negative values indicate reduced chlorophyll in seawater-treated compared to control plants receiving freshwater throughout the experiment. Effect sizes with 95% confidence intervals that do not overlap with zero are significant

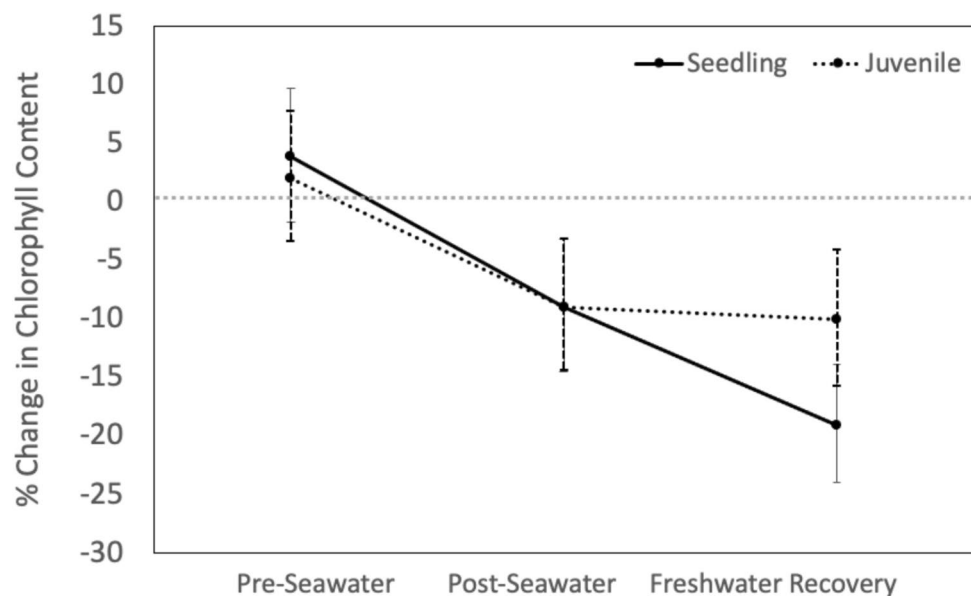


Fig. 3 Mean effect sizes for seawater effects on stomatal conductance rates for seedling and juvenile stages assessed before seawater treatment, after 2 weeks of seawater treatment, and following 3 weeks of freshwater recovery. Percent change in stomatal conductance is calculated from log-response ratios, with bias-corrected 95% confidence intervals. Negative values indicate reduced stomatal conductance in seawater-treated compared to control plants receiving freshwater throughout the experiment. Effect sizes with 95% confidence intervals that do not overlap with zero are significant

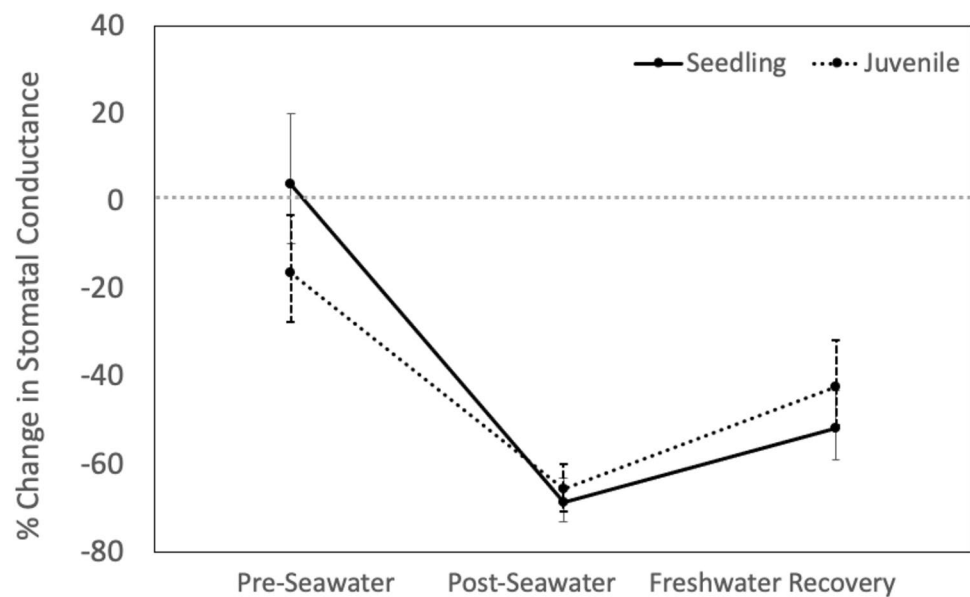


Table 3 Results from linear regression models to identify traits linked to salinity tolerance

Trait/time	Seedling	Juvenile
Chlorophyll		
Pre-seawater	0.0599	0.1837
Post-seawater	0.6091***	0.364**
Freshwater recovery	0.3218*	0.1919
Stomatal conductance		
Pre-seawater	0.126	0.0182
Post-seawater	0.3177*	0.3675*
Freshwater recovery	0.2038	0.0859

Phenotypic plasticity in leaf chlorophyll content and stomatal conductance following seawater treatment and freshwater recovery were analyzed as predictors of biomass tolerance, quantified as log-response ratio effect sizes. Sample sizes differed between seedling ($n = 12$ species/populations) and juvenile ($n = 11$ species) stages.

Regression coefficients (R^2) are reported, and significance is given as ***($P < 0.01$), **($P < 0.05$), *($P < 0.07$)

freshwater control conditions. Identification of *S. portulacastrum* as a halophyte corroborates the previous research on other populations across its global distribution (Chen et al. 2022; Messedi et al. 2004; Ramani et al. 2006). In addition, the endemic *Chenopodium oahuense* grew more under seawater treatment in populations from Maui, although the Kaua'i population grew less under seawater treatment. Thus, *C. oahuense* is conditionally halophytic, and the evolutionary drivers underlying this intraspecific population variation warrant future research. Future experiments could further investigate these species for insights into the halophytic strategies they use to thrive under elevated salinity. All other species tested had reduced growth under seawater treatment,

ranging from 3 to 95% less total biomass than control plants. Similarly, low and variable seawater tolerance was previously demonstrated at the seed-germination stage for an overlapping pool of Hawaiian coastal plant species (Walsh et al. 2023), confirming the vulnerability of these plant communities to sea-level rise. Sensitivity in young ontogenetic stages, such as that shown here for seedling and juvenile plants, may be easy to overlook in the field because of their ephemeral nature and small size. Yet, reduced germination and seedling establishment can be major drivers of population declines (Del Vecchio et al. 2020; Harper 1977), emphasizing the importance of considering salinity tolerance in early ontogeny and warning of the likely loss of biodiversity in coastal dune ecosystems under sea-level rise in Hawai'i.

In contrast to growth results, coastal dune plants generally demonstrated relatively high survival in these experimental conditions. High survival but low growth under diluted artificial seawater (20 ppt) may indicate some degree of salinity tolerance in the studied dune species, which is consistent with predictions for coastal plants growing in proximity to the ocean (Hesp 1991). Yet, because salinity has dosage-dependent effects, even reducing fitness in halophytes at full seawater strength (32 ppt) (Messedi et al. 2004; Yin et al. 2018), the fitness responses reported here are likely to be conservative compared to those that will occur in plants growing under natural conditions with sea-level rise and increased aridity due to climate change, although field trials examining plant responses to ambient and simulated flooding events are needed to confirm this prediction. Moreover, freshening via rainfall and human interventions could potentially alleviate elevated salinity, enhancing persistence of plant species that can survive through intermittent periods of salinity stress (Middleton and Boudell 2023).

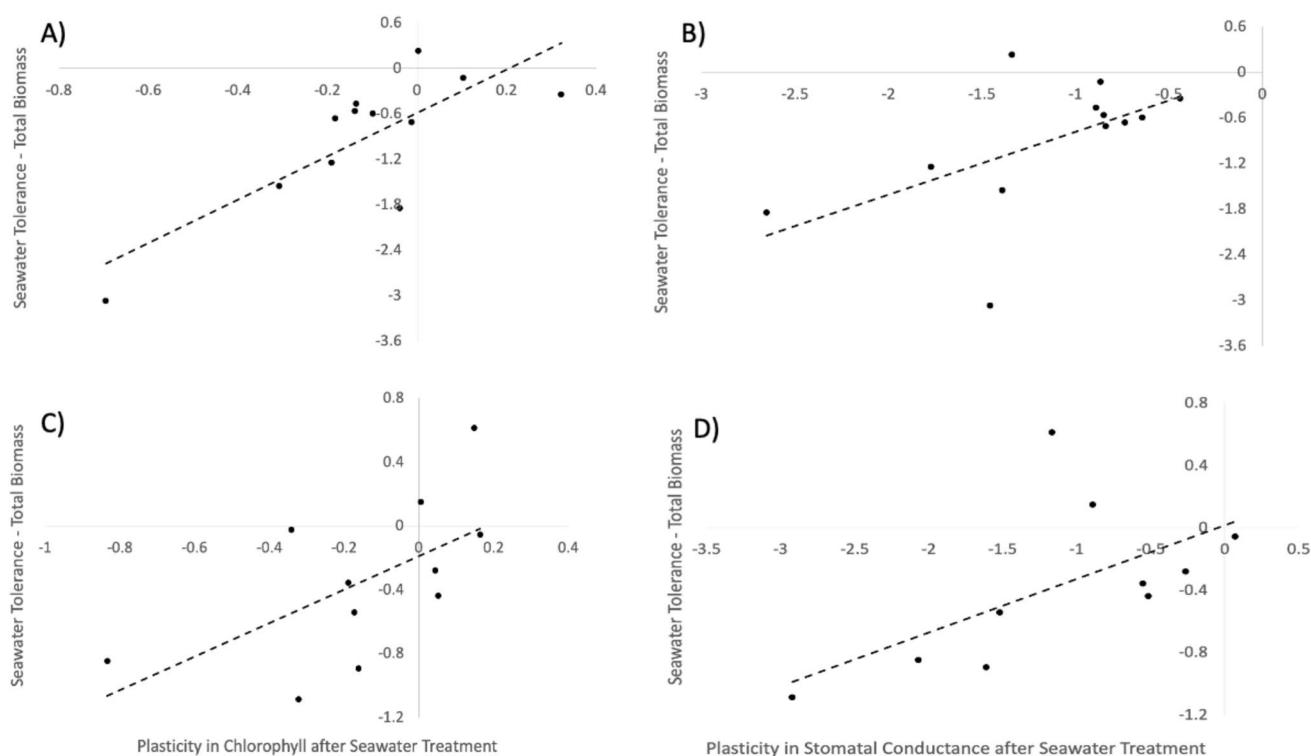


Fig. 4 Relationships between phenotypic plasticity in photosynthetic traits following seawater treatment and salinity tolerance, assessed via log-response ratios. Negative values represent decreases in seawater-treated plants compared to freshwater control plants. Significant rela-

tionships were detected for chlorophyll content in seedlings (A) and juvenile plants (C), as well as stomatal conductance in seedlings (B) and juvenile plants (D)

Variable estimates of tolerance between different metrics of fitness (e.g., survival and growth) highlight the value of a holistic assessment of fitness in the study of ecological stress tolerance, defined as the maintenance of fitness in stressed versus control conditions (Simms 2000). Fitness metrics are likely to be hierarchical, because tolerance measured as growth only matters if plants can first survive the stress. In response to seawater treatment, the lowest tolerance was thus observed for species suffering 100% mortality: the native *I. indica* (juvenile stage), *P. zeylanica* (seedling stage), and *W. indica* (both stages), as well as invasive *B. pilosa* (both stages). Moreover, for those few *I. indica* seedlings that survived seawater treatment, they had the greatest reduction in growth (95%), confirming this species to have very low salinity tolerance. In addition, the native *V. rotundifolia* and *C. subcordata* had very low tolerance in terms of survival and growth at the seedling stage, but demonstrated high survival and intermediate growth responses at the juvenile stage, illustrating consistent patterns between fitness metrics even with variable responses across ontogeny. Such ontogenetic increases in salinity tolerance have been detected in other coastal dune species, emphasizing the vulnerability of the earliest ontogenetic stages to increasing salinity stress (Alpha et al. 1996; Lum and Barton 2020). Clearly,

combining survival and growth fitness estimates improves our understanding of salinity tolerance in these dune species.

Invasive plants are a major threat to Hawai'i's native species, as they are to island floras globally (Fernández-Palacios et al. 2021; Kueffer et al. 2010). While most lowland habitats suffer the highest land-use change and invasive species abundances (Jacobi et al. 2017), Hawai'i's coastal dunes still have high densities of native plants (Warshauer et al. 2009). Identifying factors threatening native coastal species persistence is thus a research priority to conserve these narrow regions of high biodiversity. Climate change can facilitate the spread of invasive species when invaders are more tolerant than native species to novel climates (Carrara et al. 2024; Hou et al. 2014). Within our pool of 19 species, we found that the invasive species did not significantly differ from native species in salinity tolerance. In fact, at the juvenile stage, native species appear to be generally more tolerant of seawater treatment than invasive species, although high species variability means that this was not a significant effect. The most common, and thus most problematic, coastal dune invaders in Hawai'i are trees (*Casuarina equisetifolia* and *Prosopis pallida*), which differ considerably in architecture from the more typical native species architecture of prostrate, climbing shrubs. Furthermore, *C. equisetifolia* and *P. pallida* alter

nutrient availability and hydrology with major impacts on coastal ecosystem function (Dudley et al. 2020; Gallaher and Merlin 2010; Hata et al. 2012). Under elevated salinity from sea-level rise, our study suggests that performance of coastal invaders is likely to be poor, particularly for *C. equisetifolia* and *Leucaena leucocephala*, which had greatly reduced growth under seawater treatment (approx. 70 and 60% reductions compared to control plants, respectively). If such low salinity tolerance translates into reduced seedling establishment, sea-level rise could potentially limit the spread of these problematic invaders. Additional field research and demographic monitoring to track the spread and abundance of these invasive species and their recruitment dynamics are needed to further investigate these patterns.

Plants have evolved diverse strategies to mitigate the consequences of high salinity on ecophysiological function, generally involving mechanisms to avoid, sequester, or secrete salt ions (Flowers and Colmer 2015). In addition to the direct negative effects of excess salt ions on cellular function, high salinity in the soil substrate causes osmotic stress to plants by reducing soil water potential (Munns 2002). Phenotypic plasticity in photosynthetic parameters reflects plant adjustments to these direct and indirect effects, with the potential to contribute to growth tolerance. For example, plants often respond to elevated salinity by reducing stomatal conductance, thereby conserving water and increasing water use efficiency (Cambrollé et al. 2011; Harris et al. 2010; Lum and Barton 2020). Indeed, all species responded to seawater treatment with a dramatic decrease in stomatal conductance at both the seedling and juvenile stages. While reduced stomatal conductance is generally perceived to be an adaptive response under osmotic stress, we observed a positive relationship between plasticity in stomatal conductance and growth tolerance. Specifically, those species that underwent the smallest reduction in stomatal conductance had the highest tolerance, indicating that while regulated stomatal conductance may conserve water, it comes at a cost for growth. Whether this relationship reflects the relatively short duration of the experiment, such that minimal stomatal regulation would eventually be a liability, leading to high mortality under extended osmotic stress, would require further experimentation. Furthermore, seawater-treated plants failed to fully recover stomatal conductance to comparable levels of the freshwater control plants, indicating a persistent effect of salinity stress on gas exchange. Such long-lasting effects from pulses of salinity stress during early ontogenetic stages have been previously reported, even leading to reproductive delays (Lum and Barton 2020). These patterns reveal the consequences of intermittent or short-term salinity stress for ecophysiological function, highlighting the predicted negative effects that increased coastal flooding under future climate change can have on coastal plants (Hanley et al. 2019).

Leaf chlorophyll content was also highly plastic in response to seawater treatment. However, reductions in chlorophyll are likely to be a symptom of stress, reflecting disrupted ion balance under elevated salinity, rather than an adaptive plastic response to maximize performance. For example, high salinity can interfere with nitrate and ammonium uptake by roots (Messedi et al. 2004), which could limit chlorophyll production in leaves. As expected, those species that maintained the highest levels of leaf chlorophyll content had the greatest tolerance to seawater treatment. Stabilizing the light reactions of photosynthesis via anti-oxidant activity and non-photochemical quenching has been implicated in salinity tolerance of diverse plant species (Chen et al. 2022; Ramani et al. 2006; Tezara et al. 2003), corroborating the result presented here on the importance of maintaining chlorophyll during seawater exposure. Future research investigating the dynamics of these photosynthetic responses, and associated mechanisms to mitigate high salinity, would provide a better understanding of the strategies by which the most tolerant species maintain high growth under elevated salinity. In turn, more detailed physiological studies could reveal why the least tolerant species are unable to survive and grow under elevated salinity, an unexpected result for coastal dune plants.

Previous studies have similarly reported low and variable salinity tolerance in native coastal dune plants, but these studies have only examined one or a few species at a given site (Ben Amor et al. 2006; Suarez 2011; Zunzunegui et al. 2021). Comparative approaches that investigate many species are particularly important for predicting how climate change will alter community composition driven by variable salinity tolerance across species. In Hawai'i's coastal dune communities, we detected extensive variability in salinity tolerance during early ontogenetic stages, ranging from highly sensitive species at risk of decline and tolerant species that may become dominant under sea-level rise. For biodiversity hotspots, such as oceanic islands, shifts in community composition often correspond to species extinctions as endemic plants and their dependent animals are not easily rescued via meta-population dynamics (Harding and McNamara 2002). In Hawai'i's coastal dunes, native species are still common despite on-going invasion threats and anthropogenic disturbances. These narrow habitats are facing extreme and rapid climate change as sea-level rise and decreasing rainfall drive elevated salinity. Despite their evolutionary history with exposure to seawater spray and occasional inundation, we found most dune species to be sensitive to seawater at the seedling and juvenile plant stages. Seawater treatment caused high mortality in some species, reduced growth in most species, and compromised photosynthesis in all species. These results emphasize the importance of considering early ontogenetic stages, which are difficult to study but essential for stable population

dynamics. Moreover, these experimental results illustrate the importance of direct tests for stress tolerance under novel climate regimes, as we have shown dune plants growing in close proximity to the ocean to be surprisingly sensitive to seawater exposure.

Conservation efforts that mitigate salinity stress via irrigation or assisted migration inland may thus be necessary to enhance the survival and growth of sensitive coastal dune plant species under future climate change (Aitken and Whitlock 2013; Middleton and Boudell 2023). In addition, outplanting juvenile plants, which tend to have greater salinity tolerance than seedlings, may help bypass the most vulnerable seedling stage, thereby enhancing restoration success. Ensuring the persistence of coastal dune plants is critical for the maintenance of these diverse ecosystems and the services they provide, and salinity stress should be a major threat addressed in coastal dune conservation plans.

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Author contribution statement KEB conceived and designed the experiments. TS, CD, DW, and SKW collected seeds of the focal species. AH-M conducted the experiments and analyzed the data with KEB. KEB and AH-M wrote the manuscript, and all authors contributed input and edits.

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Code availability Code will be deposited with the data in DRYAD.

Declarations

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Consent to participate N/A

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